



# Smart Notary

*Decentralized Document Notarization on Polygon Blockchain*

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Kutaisi International University · 2025

## THE PROBLEM

# *"Who do you trust to prove your document is real?"*

Traditional notarization has existed for centuries — but it hasn't changed. You walk into an office, hand your document to a stranger, and trust that their stamp means something.

- A notary can be corrupt or make mistakes
- Physical stamps and signatures can be forged
- Records can be lost, altered, or destroyed
- Cross-border notarization takes days and costs money

*There has to be a better way.*



## THE SOLUTION

# How It Works

SmartNotary replaces the middleman with cryptographic proof.

### STEP 01 — HASH

## Fingerprint

When you upload a document, the system calculates its SHA-256 fingerprint. Change even one character and the hash changes completely — making tampering mathematically detectable.

### STEP 02 — STORE

## IPFS

The original file is uploaded to IPFS — a decentralized peer-to-peer storage network. No single entity controls it. Nobody can take it down.

### STEP 03 — NOTARIZE

## Blockchain

The hash, your wallet address, and a timestamp are written permanently to the Polygon blockchain via a smart contract. Immutable. Public. Forever.

*No middleman. No single point of failure. No trust required.*

ARCHITECTURE

# Tech Stack

|                |               |                                             |
|----------------|---------------|---------------------------------------------|
| FRONTEND       | React + Vite  | Fast, modern UI with Tailwind CSS           |
| AUTH           | Web3Auth      | Login with Google — no crypto needed        |
| SMART CONTRACT | Solidity      | Deployed on Polygon — low fees, fast        |
| STORAGE        | IPFS · Pinata | Decentralized, permanent file storage       |
| BACKEND        | Node.js       | Gas station — funds new users automatically |



# The Smart Contract

The entire notarization logic is 30 lines of Solidity.

```
function notarize(string memory _fileHash, string memory _ipfsCID) public {  
    require(documents[_fileHash].timestamp == 0, "Already notarized");  
    documents[_fileHash] = Document(  
        _fileHash, _ipfsCID, msg.sender, block.timestamp  
    );  
    emit DocumentNotarized(_fileHash, _ipfsCID, msg.sender, block.timestamp);  
}
```

*First registration wins. Nobody — not even the contract owner — can overwrite or delete a record. The blockchain enforces this permanently.*

USERS: LAST 7 DAYS USING MEDIAN ▾



LOAD TIME VS BOUNCE RATE

OPTIONS

START RENDER VS BOUNCE RATE

OPTIONS

DEMONSTRATION

# Live Demo

[smartnotary.vercel.app](https://smartnotary.vercel.app)

01

## Login

Google account → wallet created automatically, no MetaMask needed

02

## Upload

Select any document → SHA-256 hash generated instantly in browser

03

## Notarize

File uploads to IPFS, then transaction is written to Polygon blockchain

04

## Confirm

Transaction confirmed → permanent PolygonScan link generated

05

## Verify

Upload same file → blockchain record returned instantly

PAGE VIEWS VS ONLOAD

Page Load (LUX)

0.7s

1s

0.8s

0.6s

Page Views (LUX)

2.7Mpvs

Bounce Rate (LUX)

40.6%

OPTIONS

500K 100%

400K 80%

300K 60%

SESSIONS

Sessions (LUX)

479K

4 pvs

3.2 pvs

2.4 pvs

Session Length (LUX)

17min

PVs Per Session (LUX)

2pvs

OPTIONS

100K 40 min

80K 32 min

60K 24 min



# Proof Without a Middleman



Upload the original document → system hashes it and queries the blockchain → owner address, timestamp, and IPFS link returned instantly.



Upload a modified version → the hash is completely different → verification fails immediately. Even changing one word is detectable.

*"You cannot fake a SHA-256 hash.  
The math is the notary."*

IMPACT

# By The Numbers

0.01

POL per notarization  
Less than \$0.01 USD

∞

Record lifetime  
Permanent on blockchain

<3s

Verification time  
Instant blockchain query

0

Middlemen required  
Fully trustless system

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*The entire system costs less to run for a year than a single traditional notarization appointment. Every record is verifiable by anyone in the world, for free, forever.*





# Thank You

*SmartNotary is live and fully functional.*

[smartnotary.vercel.app](https://smartnotary.vercel.app)

REACT · SOLIDITY · POLYGON · IPFS · WEB3AUTH